DO-178C SOFTWARE COMPLIANCE REFERENCE DOCUMENT

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1. SOFTWARE REQUIREMENTS STANDARDS

All software requirements shall be clearly defined, complete, and unambiguous.
Each requirement shall be uniquely identified and documented.
Software requirements shall be accurate and internally consistent.
All requirements shall be verifiable through test, analysis, or inspection.
Software requirements shall conform to defined software requirement standards.

Traceability shall be maintained between system requirements and software requirements.
Bidirectional traceability shall exist between:
- High-level requirements (HLR)
- Low-level requirements (LLR)
- Test cases

All derived requirements shall be identified and validated.

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2. REQUIREMENTS VERIFICATION OBJECTIVES

Each software requirement shall be reviewed for:
- Accuracy
- Consistency
- Completeness
- Verifiability

Verification shall confirm that:
- Requirements correctly implement system requirements
- Requirements are testable
- Requirements are free from ambiguity

Trace data shall demonstrate mapping between system-level and software-level requirements.

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3. SOFTWARE DESIGN AND ARCHITECTURE

Software architecture shall be defined and documented.
The design shall implement all high-level requirements.
Low-level requirements shall satisfy high-level requirements.

Software design shall be:
- Accurate
- Consistent
- Complete

Software architecture shall conform to design standards.
Partitioning and modularity shall ensure isolation and fault containment.
All components shall have clearly defined interfaces.

Traceability shall ensure that all requirements are implemented in the design.

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4. SOFTWARE CODING AND IMPLEMENTATION

Source code shall comply with low-level requirements.
Source code shall conform to coding standards.
Source code shall be readable, maintainable, and verifiable.

Each code component shall be traceable to its corresponding requirement.
All variables, interfaces, and control flows shall be clearly defined.

Integration shall ensure that all software components interact correctly.
Outputs of integration shall be complete and correct.

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5. SOFTWARE VERIFICATION AND TESTING

Verification shall include:
- Reviews
- Analysis
- Testing

Test procedures shall be correct, complete, and repeatable.
Test results shall be accurate and discrepancies shall be documented.

Test coverage shall demonstrate:
- Coverage of high-level requirements
- Coverage of low-level requirements

Structural coverage shall be achieved, including:
- Statement coverage
- Decision coverage (as applicable)

Data coupling and control coupling between components shall be verified.

All anomalies detected during testing shall be recorded and resolved.

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6. TRACEABILITY REQUIREMENTS

Traceability shall be established and maintained across all lifecycle stages.

The following traceability links shall exist:
- System Requirements → Software Requirements
- High-Level Requirements → Low-Level Requirements
- Requirements → Design
- Requirements → Code
- Requirements → Test Cases

Traceability data shall be documented and verifiable.

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7. CONFIGURATION MANAGEMENT

All software artifacts shall be under configuration control.

Configuration management shall ensure:
- Version control of all artifacts
- Change tracking and traceability
- Baseline identification

Each release shall have a defined baseline.

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8. SOFTWARE QUALITY ASSURANCE

Quality assurance activities shall be independent of development.

SQA shall ensure:
- Compliance with DO-178C objectives
- Adherence to plans and standards
- Correctness of processes

Regular reviews and audits shall be conducted.

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9. SAFETY AND RELIABILITY

Software shall implement fail-safe mechanisms.
Fault detection and handling shall be defined.

Redundancy shall be used where required for safety-critical functions.

Error logging, detection, and recovery shall be implemented.

System behavior under failure conditions shall be predictable and safe.

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10. TEST COVERAGE AND VALIDATION

Test coverage shall demonstrate that all requirements are exercised.

Verification shall confirm:
- Correct execution of functionality
- Proper handling of edge cases
- Stability under expected operating conditions

Coverage gaps shall be analyzed and resolved.

END OF DOCUMENT
